KING SALMAN INTERNATIONAL UNIVERSITY

Field of Technological Industries

All Program

Course : Mathematics 1 (MAT 111)

Lecture 2: Function

Dr : Ahmed Ali

Course contents

Calculus part

- Functions
- Limits
- Continuity
- Derivatives
- Applications of differentiation

Linear Algebra part

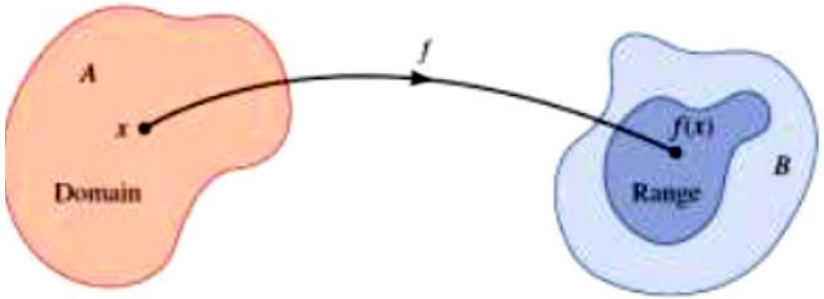
- Systems of Linear Equations
- Theory of Matrices
- Applications in Systems of Equations

textbook: R. Smith, R. Minton, z. Rafhi, Calculus: Early Transcendental Functions, 5th Edition, McGraw-Hill Education.

Functions

The word ***function*** is often used to describe the relation between two or more variables or to show the dependence of one variable on the other variables.

Definition: A function f from a set A to a set B is a rule that assigns to each element x in A one and only one element $y = f(x)$ in B. The set A is called the domain of the function, and the set of assigned elements in B is called the range.

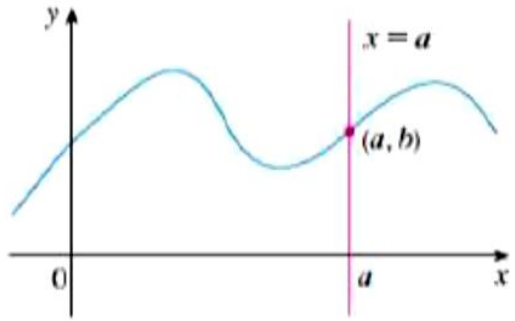


Think: The function f is a machine, that takes input x and produces a corresponding output $y = f(x)$.

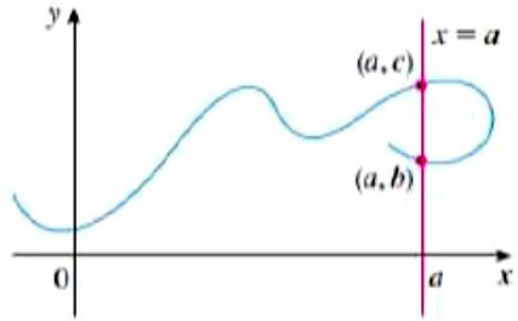


Graphically: The vertical Line Test

A curve in the $xy-$ plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.



A function



A graph not a function

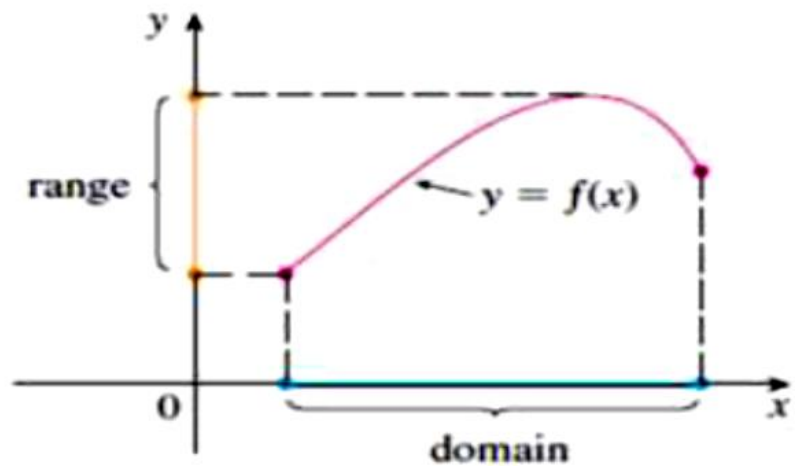
Domain and Range

Domain(f) = the set of all possible input values "x"that we can substitute in the given function $f(x)$.

Domain(f) =the projection of the curve of f on the x-axis.

Range(f) = the set of all output values "y" we obtained by substituting the domain values "x" in the function $f(x)$.

Range(f) = the projection of the curve of f on the y-axis.

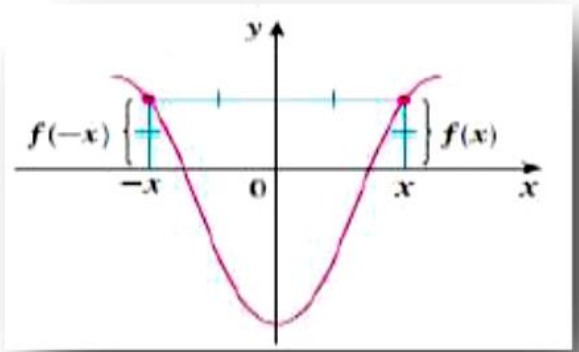


- x is called the independent variable.
- y is called the dependent variable.

Symmetry

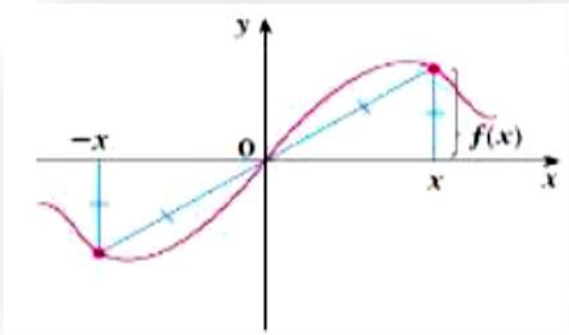
Even Function

If a function f satisfies $f(-x) = f(x)$ for every number x
In its domain, then f is called an even function.
(graph of f is symmetric with respect to the y-axis)



Odd Function

If a function f satisfies $f(-x) = -f(x)$ for every number x
In its domain, then f is called an odd function.
(graph of f is symmetric about the origin)



Algebraic and absolute value functions

The ***functions*** can be classified into algebraic, piecewise, and transcendental functions. All polynomials, radicals, rational, and power ***functions*** are algebraic.

Polynomial Functions

A function of the form $\boldsymbol{f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}$,

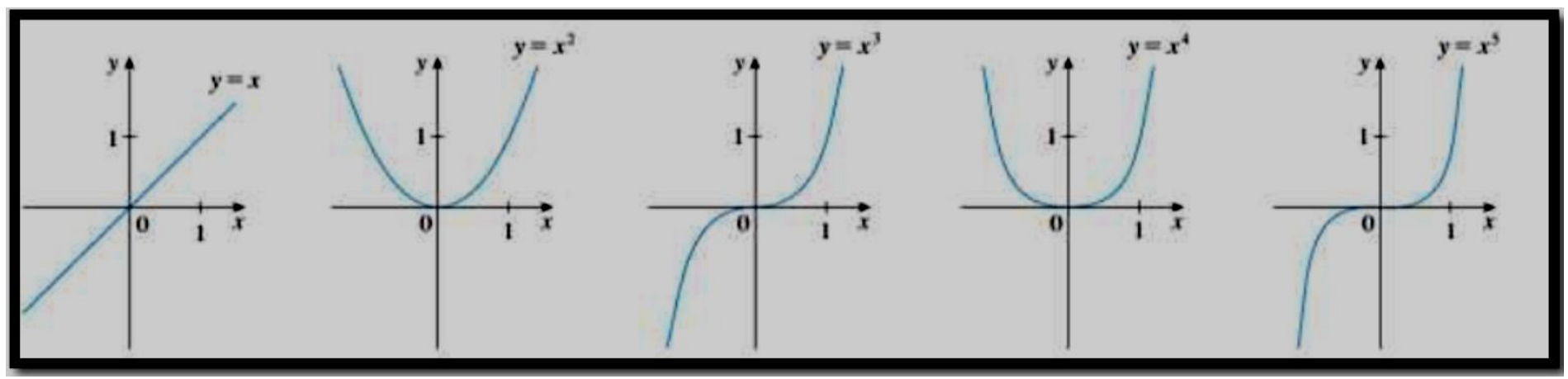
where $a_0, a_1, \ldots, a_n$ are real numbers (the coefficients of the polynomial with $a_n \neq 0$) and $n \geq 0$ is an integer, is a polynomial function of degree n.

Examples: **Constant function** $f(x) = 4$ (polynomial of degree zero),

Linear function $f(x) = 6x + 1$ (polynomial of degree one),

Quadratic function(parabolas) $f(x) = 2x^2 + 7x - 2$ (polynomial of degree two).

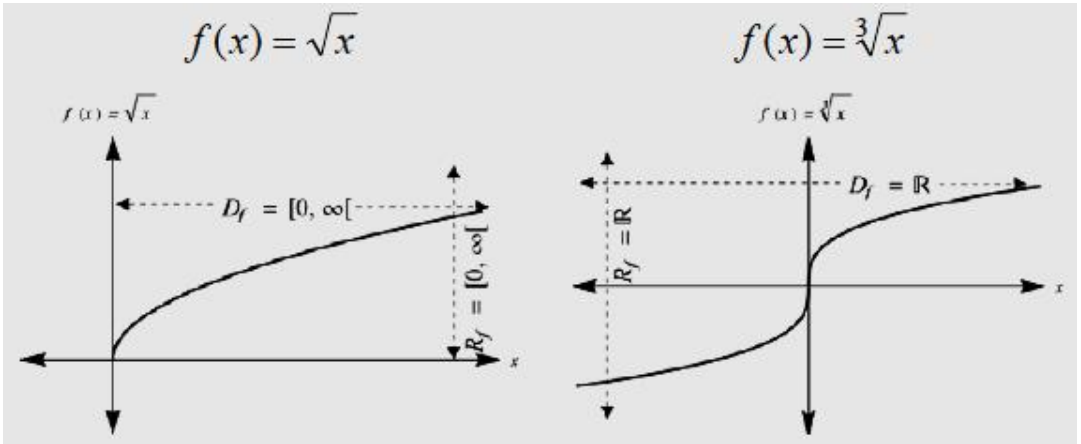
Note: The domain of any polynomial is $\mathbb{R} =]-\infty, \infty[$.

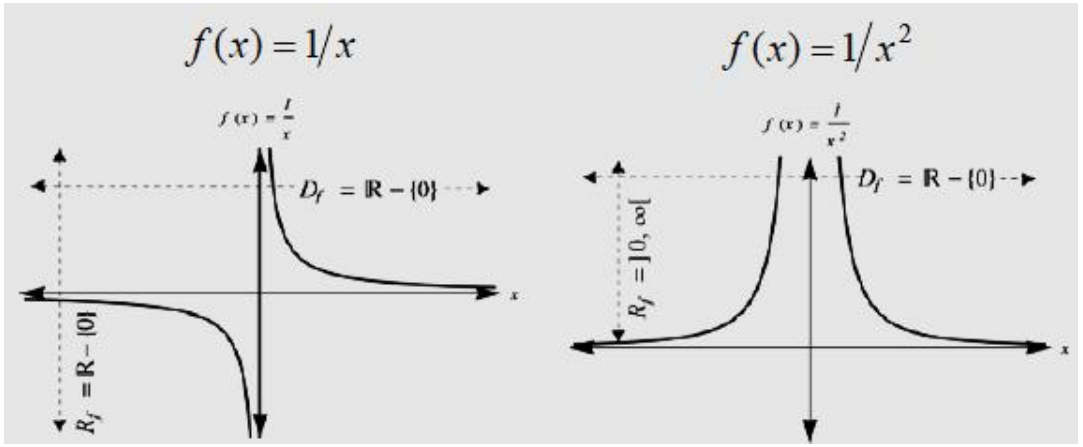


Definitions (*Radical, rational, and power functions*)

Let $P(x)$ and $Q(x)$ be polynomials and n is a positive integer.

- The radical is the ***function*** $f(x)=\sqrt[n]{P(x)}$
- The ***domain*** of radical is $\mathbb{R}$ if n is odd.
- The ***domain*** of radical is $\{x : P(x) \geq 0\}$ if n is even.
- The rational is the ***function*** $f(x)=P(x)/Q(x)$.
- The ***domain*** of a rational ***function*** is $\mathbb{R}-\{x : Q(x)=0\}$.
- The power ***function*** is $f(x)=x^m$ where m is a real number .





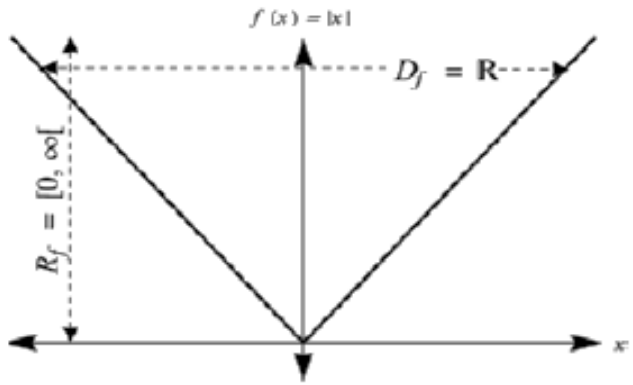
The ***function*** need not be defined by a single rule. The piecewise ***function*** is defined by multiple rules on different intervals in the ***domain*** of the ***function***.

Definition (***Absolute value function***)

The absolute value ***function*** is a piecewise one defined as

$$f(x) = |x| = \begin{cases} x & if \ x \geq 0 \\ -x & if \ x < 0 \end{cases}$$

The ***domain*** of the absolute value ***function*** is $\mathbb{R}$ as shown.



Example 1: Find the ***domain*** and the ***range*** of the following ***functions***:

i. $f(x)=\sqrt{x-2}$ ii. $g(x)=\sqrt[3]{2-x}$

iii. $h(x)=\dfrac{3x+2}{x-2}$ iv. $p(x)=x^2+4$

Solution

i. $f(x)=\sqrt{x-2} \xrightarrow{D_f} x-2\geq 0 \Rightarrow x\geq 2 \Rightarrow \boxed{D_f=[2,\infty[}$

$\because x-2\geq 0 \Rightarrow \sqrt{x-2}\geq 0 \xrightarrow{y=f(x)} y\geq 0 \Rightarrow \boxed{R_f=[0,\infty[}$

ii. $g(x)=\sqrt[3]{2-x} \xrightarrow{D_f} \boxed{D_f=\mathbb{R}} \Rightarrow \boxed{R_f=\mathbb{R}}$

iii. $h(x)=\dfrac{3x+2}{x-2} \xrightarrow{D_f} x-2\neq 0 \Rightarrow x\neq 2 \Rightarrow \boxed{D_f=\mathbb{R}-\{2\}}$

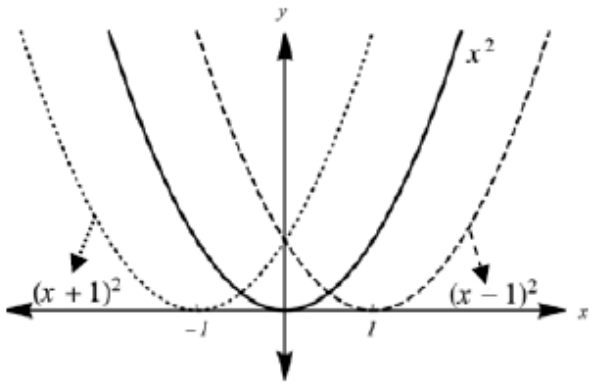
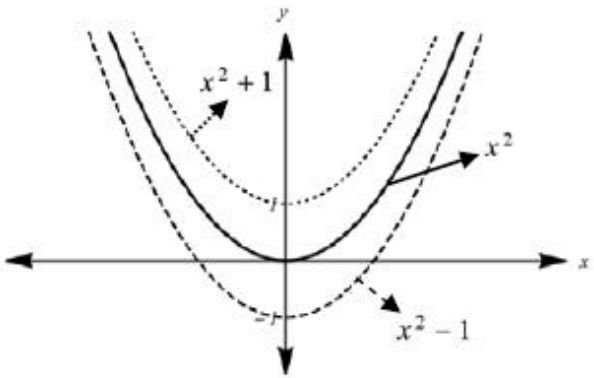
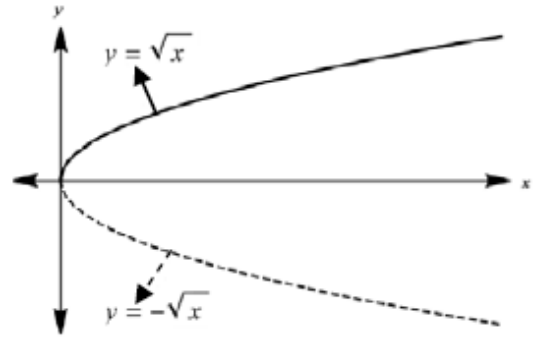
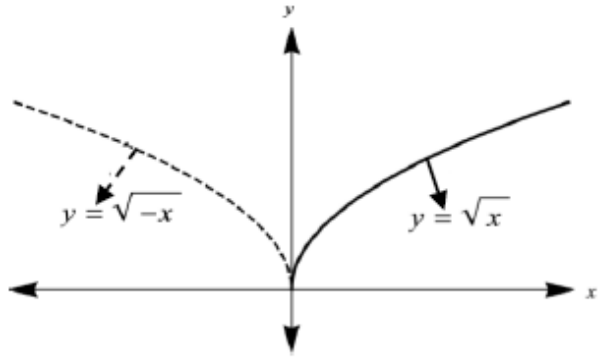
$y=\dfrac{3x+2}{x-2} \Rightarrow xy-2y=3x+2 \Rightarrow x(y-3)=2y+2 \Rightarrow$

$x=\dfrac{2y+2}{y-3} \Rightarrow y\neq 3 \Rightarrow \boxed{R_f=\mathbb{R}-\{3\}}$

iv. $p(x)=x^2+4 \xrightarrow{D_f} \boxed{D_f=\mathbb{R}}$

$\because x^2\geq 0 \Rightarrow x^2+4\geq 4 \xrightarrow{y=p(x)} y\geq 4 \Rightarrow \boxed{R_f=[4,\infty[}$

The ***graph*** of ***function*** can be used to determine the ***domain*** and the ***range*** of ***function***. The ***graph*** of ***function*** can be shifted or reflected as shown in **Tables**

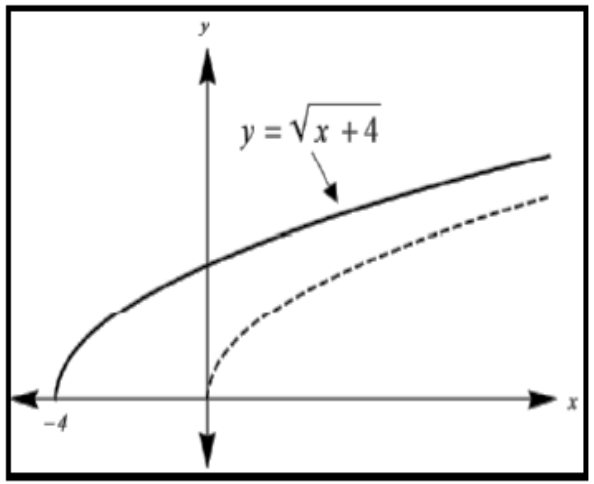
Expression/Action	**Example**
$y = f(x \pm k)$ Shift k units to the left (+) or to the right (-).	$y = (x \pm 1)^2$ 
$y = f(x) \pm k$ Shift k units upward (+) or downward (-).	$y = x^2 \pm 1$ 
$y = -f(x)$ Reflection about $x-$axis	$y = -\sqrt{x}$ 
$y = f(-x)$ Reflection about $y-$axis	$y = \sqrt{-x}$ 

Example 2: Sketch the ***graph*** of $f(x)$ then find the ***domain*** and the ***range*** of $f(x)$ if:

i. $f(x)=\sqrt{x+4}$ ii. $f(x)=|x|/x$

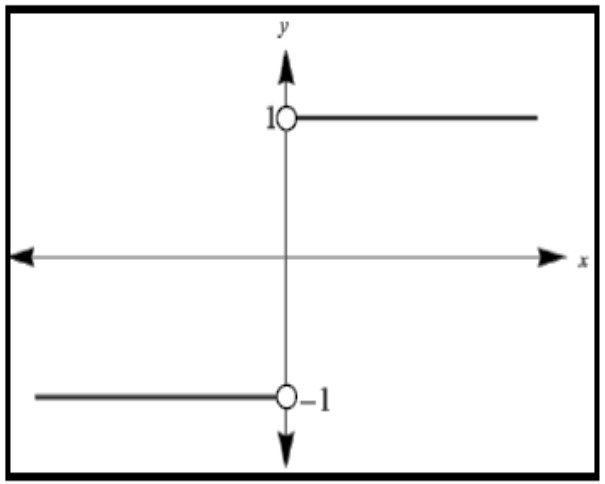
Solution

i. $f(x)=\sqrt{x+4} \Rightarrow \sqrt{x}$ is shifted 4 units to the left.



From the ***graph***, the ***domain*** of $f(x)$ is $\boxed{x\in[-4,\infty[}$ and the ***range*** of $f(x)$ is $\boxed{y\in[0,\infty[}$

ii. $f(x)=\dfrac{|x|}{x}=\begin{cases} x/x & if\ x>0 \\ -x/x & if\ x<0 \end{cases}=\begin{cases} 1 & if\ x>0 \\ -1 & if\ x<0 \end{cases}$



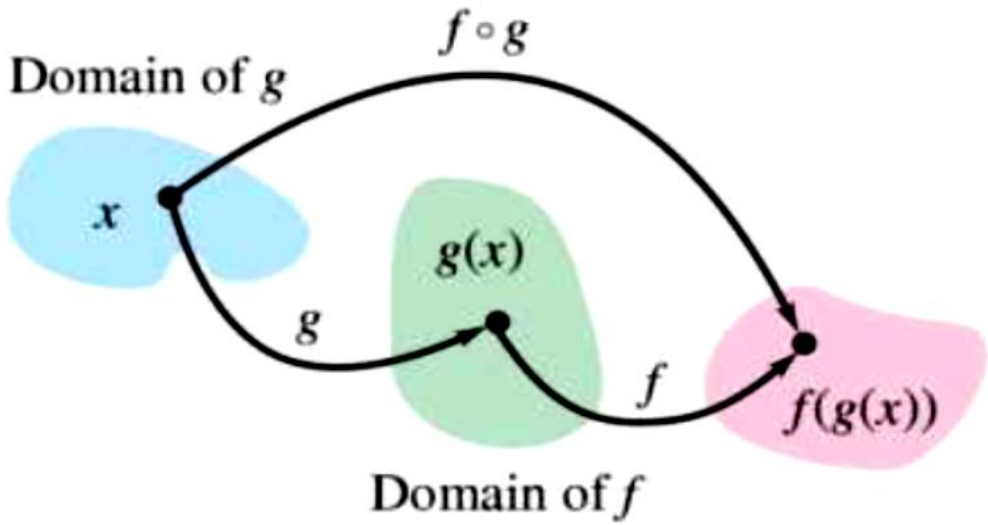
From the ***graph***, the ***domain*** of $f(x)$ is $\boxed{x\in\mathbb{R}-\{0\}}$ and the ***range*** of $f(x)$ is $\boxed{y\in\{-1,1\}}$

Composition of functions

DEFINITION: If f and g are functions, the composite function $(f \circ g)$ ("f composed with g") is defined by

$$(f \circ g)(x) = f\big(g(x)\big).$$

The domain of $(f \circ g)$ consists of the numbers x in the domain of g for which $g(x)$ lies in the domain of f.



The domain of the composite function $f \circ g$

Example: If $f(x) = \sqrt{x}$ and $g(x) = x + 1$, find

(a) $(f \circ g)(x)$ **(b)** $(g \circ f)(x)$ **(c)** $(f \circ f)(x)$ **(d)** $(g \circ g)(x)$

Composite	Domain
$(f \circ g)(x) = f\big(g(x)\big) = \sqrt{g(x)} = \sqrt{x+1}$	$[-1, \infty[$
$(g \circ f)(x) = g\big(f(x)\big) = f(x) + 1 = \sqrt{x} + 1$	$[0, \infty[$
$(f \circ f)(x) = f\big(f(x)\big) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{1/4}$	$[0, \infty[$
$(g \circ g)(x) = g\big(g(x)\big) = g(x) + 1 = x + 2$	$]-\infty, \infty[$

➢ **Note that:** the functions $g \circ f$ and $f \circ g$ are usually quite different.

Inverse functions:

If $y = f(x)$ is 1-1 function, then there exist a unique inverse function $f^{-1}(x)$ with the following properties

Properties of the inverse functions

1- The domain of f^{-1} = range of f.

2- The range of f^{-1} = domain of f.

3- $y = f(x)$ is equivalent to $x = f(y)$.

4- The graph of f and f^{-1} are reflections in the line $y = x$.

5- $f\left(f^{-1}(x)\right) = x$ for x in the domain of f^{-1}.

6- $f^{-1}\left(f(x)\right) = x$ for x in the domain of f.

➢ **CAUTION** The symbol $f^{-1}(x)$ does not mean $\frac{1}{f(x)}$.

To get $f^{-1}(x)$:

Algebrically

a) Let $y = f(x)$,
b) Write x in terms of y,
c) Interchange x and y,
d) Replace y by $f^{-1}(x)$.

Geometrically

Reflecting the graph of $y = f(x)$ in the line $y = x$.

Example: Find the inverse function $f^{-1}(x)$ when $f(x) = 5x^2 + 4,\ x \in [0, \infty[$. Find the domain and range of the inverse function.

Solution

We write, $y = 5x^2 + 4$

$$\Longrightarrow \quad x^2 = \frac{y-4}{5}$$

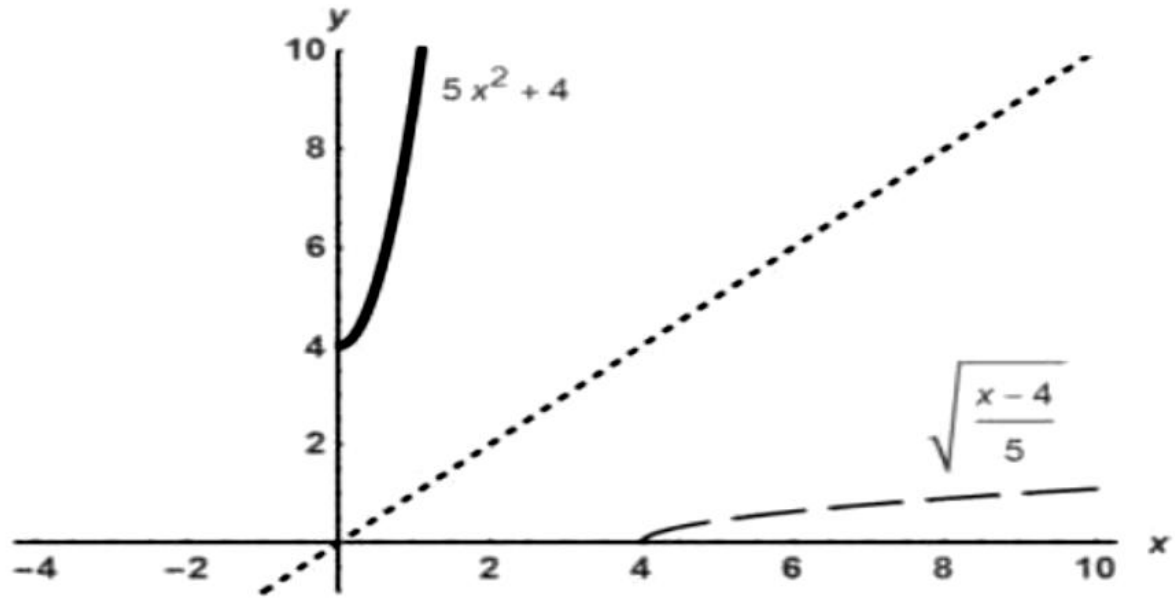
$$\Longrightarrow \quad x = +\sqrt{\frac{y-4}{5}}$$

Interchanging x and y

$$\Longrightarrow \quad y = \sqrt{\frac{x-4}{5}}$$

Replace y by $f^{-1}(x)$

$$\Longrightarrow \quad f^{-1}(x) = \sqrt{\frac{x-4}{5}}$$



The domain of $f(x)$ is $x \in [\,0, \infty\,[$ and the range of $f(x)$ is $y \in [\,4, \infty\,[$

So, the domain of $f^{-1}(x)$ is $x \in [\,4, \infty\,[$ and the range of $f^{-1}(x)$ is $y \in [\,0, \infty\,[$.

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THANK YOU

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